

AI-Powered Worship Presentation System

Full Product Requirements Document

1. Core Goal

Develop a Windows desktop worship presentation application that fully replaces traditional systems while integrating AI-driven sermon transcription and scripture detection.

2. Core Features

The system must include the following essential features:

- Song Management System: Create, edit, delete, search, and organize songs with verse/chorus structure.
- Scripture System: Local Bible storage, multi-version support, and fast lookup.
- Presentation Builder: Create slides with text and images; drag-and-drop arrangement.
- Live Presentation Mode: Dual-screen display with preview and live output.
- Media Support: Playback of video, images, and audio.
- Service Manager: Organize full service flow with quick switching between items.

3. AI Features

- Live Transcription: Real-time conversion of sermon speech into text.
- Scripture Detection: Automatic identification of Bible references from speech.
- Smart Projection Button: Projects the most recently detected scripture.
- Operational Modes: Manual Mode, Smart Mode, and Auto Mode.

4. Tech Stack

- Frontend: Electron + React + TypeScript
- Backend: Node.js + NestJS
- AI Speech Recognition: OpenAI Whisper (primary), Vosk (offline fallback)
- Database: SQLite (local), PostgreSQL (optional cloud sync)
- Real-Time Communication: WebSockets
- Packaging: Electron Builder (.exe installer)

5. System Architecture

Microphone Input → Speech-to-Text → Transcription Stream → Scripture Detection Engine → Backend → WebSocket Events → UI → Projector Output

6. How It Works

- Operator loads service plan.
- Preacher speaks and system transcribes in real-time.
- Detected scriptures are identified and stored.
- Operator clicks 'Project Scripture'.
- System displays the verse instantly on screen.

7. UI Requirements

- Left Panel: Library (Songs, Scriptures, Media)
- Center: Preview Screen
- Right Panel: Live Transcript and Detected Scriptures
- Top Controls: Next, Previous, Clear, Project Scripture

8. Performance Requirements

- Transcription latency under 2 seconds.
- Slide transitions under 200 milliseconds.
- High stability during live usage.
- Offline capability for core features.

9. Security Requirements

- Local-first data storage.
- Optional cloud synchronization.
- Encryption for sensitive configurations.

10. Development Phases

- Phase 1: Basic presentation system.
- Phase 2: Song and scripture management.
- Phase 3: Dual-screen live mode.
- Phase 4: Add transcription.
- Phase 5: Add scripture detection.
- Phase 6: Implement smart projection feature.

